

SEBASTIAN CORREA-GALLEGO

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Research Interests

Evolution and ecology are not subfields of biology — they are its organizing logic, present from the emergence of cellular complexity to the structure of communities in extreme environments. My research interests sit at this intersection: I am drawn to microbial systems as tractable windows into how life assembles, persists, and transforms under ecological and energetic constraint. This includes questions about community organization along environmental gradients, the physiological and molecular strategies that underlie ecological success, and the eco-evolutionary dynamics that connect cellular function to community-level patterns. I am also interested in the quantitative and computational frameworks that make these questions formally tractable, and in connecting field-based inquiry with analytical rigor across biological scales.

Education

Universidad EAFIT, B.Sc. in Biology 2022–2026 (*expected*)

- Undergraduate researcher in microbial ecology and evolution.
- Thesis: cultivable microbial community structure across a light gradient in an Andean volcanoclastic cave.

Research Experience

Visiting Student Intern Aug 2025–Jan 2026

ECISO Lab, Department of Biological Sciences, Purdue University

- Conducted experimental work in evolutionary and cellular biology using *S. cerevisiae* as a model system, including high-resolution growth curve acquisition, cell harvesting under defined carbon regimes, and kinetic data analysis.
- Contributed to proteomics-based workflows, including sample preparation and downstream integrative analysis across molecular and cellular scales.

Undergraduate Thesis Researcher 2024–2026

Universidad EAFIT · Advisor: Prof. Nicolás Pinel Peláez, Ph.D.

Thesis: *Cultivable Microbial Community Structure Along a Light Gradient in a Tropical Volcanoclastic Cave*

Research Monitor Jul 2024–Dec 2024

Universidad EAFIT

- Contributed to the project *Positioning of Microorganisms as Fundamental Actors for the Maintenance of Andean Forests*, supporting project development through scientific systematization, written reporting, and research communication.

Academic Service

Student Director, Research Group on Microbiology and Astrobiology Oct 2023–May 2025

Universidad EAFIT, affiliated with the Research Group on Geosciences and Biodiversity (GEBI)

- Coordinated a student-led research group focused on microbiology, astrobiology, scientific discussion, and undergraduate research initiatives.

Conferences and Presentations

Research Proposal Presentation 2024

2nd Symposium of Biology, Universidad EAFIT (awarded Second Place)

Honors and Recognition

Visiting Student Intern, Purdue University (UREP-C Program) 2025–2026

Undergraduate Scholarship, Comfama and Fundación Fraternidad Medellín 2022

Technical Skills

Laboratory and biological methods: microbial cultivation, environmental sampling, ecological field records, morphotype characterization, yeast handling, cell physiology, growth kinetics, and microscopy-based observation.

Quantitative and computational tools: R, Python, Linux/bash, $\text{\LaTeX}$, QGIS, biological data analysis, data visualization, and introductory bioinformatics workflows.

Scientific workflows: scientific writing, research communication, literature synthesis, and figure preparation.

Languages: Spanish (native), English (working proficiency).

References

Nicolás Pinel Peláez, Ph.D.

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